MODULE 5 — MOTION ANALYSIS

1. Introduction to Motion Analysis

Definition

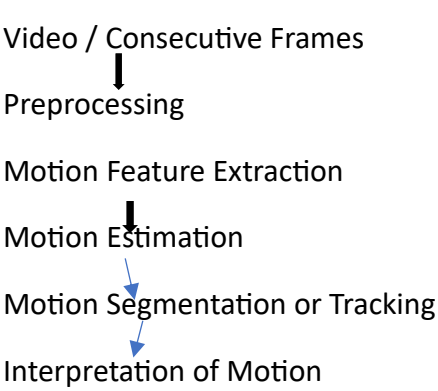
Motion analysis is the process of detecting and measuring changes in objects or image regions over time. It uses sequences of images or video frames to determine whether objects are moving, their direction and speed, and how the scene changes.

Motion information is obtained by comparing two or more frames captured at different times.

Types of motion

- Rigid motion: translation and rotation without deformation.
- Non-rigid motion: deformation, such as human facial expressions.
- Camera motion: movement of the camera itself.
- Object motion: movement of objects relative to the scene.
- Apparent motion: motion inferred from image changes; it may differ from true 3D motion.

block diagram



Advantages

- Detects moving objects and activity.
- Enables tracking over time.
- Supports behaviour and event analysis.
- Helps estimate camera and object movement.

Disadvantages

- **Sensitive to lighting changes** and shadows.

- Occlusion can interrupt tracking.
- Camera movement complicates object-motion estimation.
- Motion blur and low frame rates reduce accuracy.

Applications

Traffic monitoring, robotics, sports analysis, surveillance, gesture recognition, and autonomous navigation.

2. Background Subtraction and Modelling

Definition

Background subtraction is a technique that separates moving foreground objects from a relatively stationary background by comparing the current video frame with a model of the background.

For example, a pedestrian moving across a stationary camera's view can be detected by comparing each frame against a background estimate.

Basic mathematical model

Let $I_t(x,y)$ be the current frame and $B_t(x,y)$ the background estimate.

A simple difference is:

$$ D_t(x,y)=|I_t(x,y)-B_t(x,y)| $$

A foreground mask can be obtained by thresholding:

$$ F_t(x,y)= \begin{cases} 1,&D_t(x,y)>T\\ 0,&D_t(x,y)\leq T \end{cases} $$

Block diagram

Video Frame

Background Model

Calculate Difference from Background

Apply Threshold

Foreground Mask

Morphological Noise Removal

Moving-Object Detection

Working

1. Capture a background frame or initialise a background model.

2. Read the next video frame.
3. Compare the current frame with the background estimate.
4. Threshold the difference to obtain foreground pixels.
5. Remove small noise regions using morphological operations.
6. Group connected foreground pixels into objects.
7. Update the background model to accommodate gradual scene changes.

Common algorithms

A. Frame differencing: subtracts the current frame from a previous frame. It detects changes but may highlight only parts of moving objects.

B. Running average: updates the background gradually:

\[B_t=(1-\alpha)B_{t-1}+\alpha I_t \]

Here, \(\alpha\) is the learning rate.

C. Gaussian Mixture Model (GMM): models the background intensity of each pixel using a mixture of Gaussian distributions. It can accommodate recurring background changes, such as moving leaves.

Advantages

- Simple and effective for stationary-camera scenes.
- Supports real-time moving-object detection.
- Does not necessarily require a labelled training dataset.
- Can identify foreground regions before tracking.

Disadvantages

- Sensitive to shadows and illumination changes.
- Dynamic backgrounds can cause false detections.
- Stationary foreground objects may become part of the background.
- Camera motion can invalidate a simple background model.

Applications

Traffic surveillance, people counting, security monitoring, wildlife cameras, and industrial monitoring.

3. Optical Flow

Definition

Optical flow estimates the apparent motion of image pixels between consecutive frames. It represents motion as a vector field, where each vector indicates an estimated direction and magnitude of image displacement.

Optical flow measures apparent 2D image motion, not necessarily true 3D object motion.

Principle

The brightness constancy assumption states that a moving image point maintains approximately the same brightness over a short interval:

$$ I(x,y,t)=I(x+u,y+v,t+\Delta t) $$

Using a first-order Taylor expansion gives the optical-flow constraint:

$$ I_xu+I_yv+I_t=0 $$

where u and v are horizontal and vertical image velocities.

This is one equation with two unknowns, so additional assumptions are required.

Block diagram

Frame at Time t

Frame at Time t + 1

Calculate Spatial and Temporal Gradients

Apply Optical-Flow Method

Estimate Motion Vectors

Visualize or Track Motion

Types of optical flow

1. Sparse optical flow: estimates motion only at selected points, such as corners. It is computationally efficient.

2. Dense optical flow: estimates a motion vector for nearly every pixel. It provides detailed motion information but is usually more computationally expensive.

Lucas–Kanade optical flow

The Lucas–Kanade method assumes that neighbouring pixels have similar motion. Multiple optical-flow constraint equations are combined over a local window and solved by least squares.

\[\begin{bmatrix} \sum I_x^2&\sum I_xI_y\\ \sum I_xI_y&\sum I_y^2 \end{b matrix} \begin{bmatrix}u\\v\end{bmatrix} = -\begin{bmatrix} \sum I_xI_t\\ \sum I_Yi _t \end{bmatrix} \]

The sums are calculated over the local window.

Advantages

- Estimates direction and magnitude of apparent motion.
- Useful for tracking and motion analysis.
- Sparse methods can be efficient.
- Dense methods provide detailed motion fields.

Disadvantages

- Relies on assumptions such as brightness constancy.
- Struggles with large motion unless multi-scale methods are used.
- Textureless areas produce ambiguous estimates.
- Occlusion and motion blur can reduce accuracy.

Applications

Object tracking, video stabilization, autonomous vehicles, action recognition, sports analysis, and motion segmentation.

4. KLT — Kanade–Lucas–Tomasi Tracking

Definition

KLT is a feature-based tracking approach that detects distinctive image points and tracks them between consecutive frames using local image information. The Tomasi component is commonly used to select good features to track, while the Lucas–Kanade method estimates their displacement.

Principle

Good tracking points have strong intensity variation in two directions. A structure tensor helps distinguish these points from flat regions and edges.

\[M= \begin{bmatrix} \sum I_x^2&\sum I_xI_y\\ \sum I_xI_y&\sum I_y^2 \end{bmatrix} \]

The eigenvalues \(\lambda_1,\lambda_2\) indicate how much intensity varies in two directions. Points with a sufficiently large minimum eigenvalue are often suitable for tracking.

Block diagram

First Video Frame

Detect Good Features to Track

Calculate Feature Displacements

Search for Matching Points in Next Frame

Refine Motion Estimate

Reject Lost / Inconsistent Points

Tracked Feature Points

Working

1. Detect corners or good tracking points in the first frame.
2. Extract local image neighbourhoods.
3. Estimate their displacement in the next frame using Lucas–Kanade.
4. Refine the estimated locations iteratively.
5. Reject points that fail tracking or consistency checks.
6. Repeat across subsequent frames.

Advantages

- **Efficient for tracking selected feature points.**
- Good for small inter-frame movements.
- Can be used with pyramids to handle moderately larger displacements.
- Useful for estimating camera movement.

Disadvantages

- Points can be lost under occlusion or large motion.
- Tracking degrades when illumination changes significantly.
- Flat regions do not provide reliable features.
- Periodic re-detection may be necessary.

Applications

Video stabilization, visual odometry, object tracking, motion estimation, and augmented reality.

5. Spatio-Temporal Analysis

Definition

Spatio-temporal analysis studies how image information varies across both space and time. It combines spatial characteristics, such as shape and texture, with temporal characteristics, such as movement and duration.

A video can be represented as a three-dimensional data volume:

$$ I(x,y,t) $$

where x,y are spatial coordinates and t is time.

Types of spatio-temporal features

- Motion trajectories.
- Temporal changes in intensity.
- Motion boundaries.
- Space-time interest points.
- Motion history images.
- Changes in shape or texture across frames.

Block diagram

Video Sequence

Spatial Feature Extraction

Temporal Feature Extraction

Combine Spatial and Temporal Features

Activity / Event Classification

Output Interpretation

Working

1. Acquire consecutive video frames.
2. Extract spatial features such as edges, shapes, and texture.
3. Calculate temporal features such as optical flow or trajectories.
4. Combine features over a temporal window.
5. Identify patterns associated with actions or events.
6. Classify or interpret the observed activity.

Advantages

- Captures both appearance and movement.
- Supports recognition of complex actions.
- Can distinguish static objects from activities.
- Useful for event detection in video.

Disadvantages

- Requires multiple frames and more computation.
- Sensitive to changes in camera position and frame rate.
- Occlusion can interrupt trajectories.
- Long sequences can require substantial memory.

Applications

Human action recognition, gesture analysis, surveillance, sports analytics, medical movement analysis, and human-computer interaction.

6. Dynamic Stereo

Definition

Dynamic stereo extends stereo vision to moving scenes. It uses images from two or more cameras, potentially across time, to estimate 3D structure and movement.

Ordinary stereo estimates depth from the disparity between corresponding points in different camera views. Dynamic stereo additionally considers changes in scene geometry and point positions over time.

Stereo depth principle

For a rectified stereo camera pair:

\[Z=\frac{fB}{d} \]

where:

- \(Z\) is depth.
- \(f\) is focal length expressed in compatible image units.
- \(B\) is the baseline distance between cameras.
- \(d\) is disparity, the horizontal displacement between corresponding points.

Block diagram

Left Camera

Right Camera

Camera Calibration and Rectification

Correspondence / Disparity Estimation

3D Depth Reconstruction

Temporal Matching and 3D Motion Estimation

Working

1. Calibrate the cameras.
2. Rectify the stereo images so corresponding points lie on the same image rows.
3. Match corresponding image features.
4. Estimate disparity and calculate depth.
5. Repeat the process over time.
6. Match reconstructed 3D points across frames.
7. Estimate object or scene motion in 3D.

Advantages

- Provides depth and 3D motion information.
- Can reconstruct moving objects.
- Useful where monocular motion alone is ambiguous.

Disadvantages

- Requires reliable calibration and correspondence.
- Textureless or repetitive regions are difficult to match.
- Occlusions cause missing correspondences.
- Camera synchronization and moving objects complicate reconstruction.

Applications

Robotics, autonomous driving, motion capture, 3D surveillance, and dynamic scene reconstruction.

7. Motion Parameter Estimation

Definition

Motion parameter estimation determines the mathematical parameters that describe the movement of an object or camera between frames. Depending on the motion model, parameters may represent translation, rotation, scale, affine deformation, or 3D pose.

Common motion models

Translation:

$$ x'=x+t_x,\qquad y'=y+t_y $$

Rigid 2D transformation:

$$ \begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix} \cos\theta&-\sin\theta\\ \sin\theta&\cos\theta \end{bmatrix} \begin{bmatrix}x\\y\end{bmatrix} + \begin{bmatrix}t_x\\t_y\end{bmatrix} $$

Affine transformation:

$$ \begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}a&b\\c&d\end{bmatrix} \begin{bmatrix}x\\y\end{bmatrix} + \begin{bmatrix}t_x\\t_y\end{bmatrix} $$

Block diagram

Two or More Video Frames

Detect and Match Features

Estimate Initial Motion Parameters

Fit Motion Model

Reject Outliers (e.g., RANSAC)

Refine Parameters

Output Motion Estimate

Working

1. Detect corresponding features across frames.
2. Calculate their displacement.
3. Select a suitable model such as translation, rigid, or affine motion.
4. Estimate the parameters from matched points.
5. Use a robust method such as RANSAC to reject incorrect matches.
6. Refine the model using the inliers.
7. Evaluate the residual errors and output the final estimate.

Advantages

- Provides a compact description of motion.
- Supports video stabilization and camera tracking.
- Robust estimation methods can handle outliers.
- Enables geometric analysis of image sequences.

Disadvantages

- Accuracy depends on correct feature matching.
- A simple model may not represent non-rigid motion.
- Large occlusions or few features reduce reliability.
- Incorrect assumptions about camera motion can produce errors.

Applications

Video stabilization, camera-motion estimation, visual odometry, image registration, and robotics.

MODULE 5 — SHAPE FROM X

8. Introduction to Shape from X

Definition

Shape from X refers to a family of computer vision techniques that estimate the three-dimensional shape, surface orientation, or depth of an object from one or more visual cues.

Here, X represents a source of information, such as shading, texture, colour, motion, edges, or stereo images.

A normal image contains only two-dimensional pixel measurements. Shape-from-X techniques use assumptions about illumination, surface reflectance, camera geometry, texture, or movement to infer 3D structure.

Main techniques in your syllabus

1. Shape from shading.
2. Light at surfaces and the Phong model.
3. Reflectance map.
4. Albedo estimation.
5. Photometric stereo.

6. Surface smoothness constraints.
7. Shape from texture.
8. Shape from colour.
9. Shape from motion.
10. Shape from edges.

General block diagram

Input Image(s)

Extract Visual Cues (Shading / Texture / Colour / Motion / Edges)

Apply Geometric and Reflectance Models

Estimate Surface Normals and / or Depth

3D Shape / Surface Reconstruction

Advantages

- Recovers useful 3D information from visual data.
- Supports reconstruction without physically touching the object.
- Different cues can complement one another.
- Useful for measurement and scene understanding.

Disadvantages

- Results depend on assumptions about lighting, materials, and camera geometry.
- Some cues are ambiguous when used alone.
- Occlusion and noise can make reconstruction difficult.
- Accurate 3D reconstruction may require calibration or multiple views.

Applications

Robotics, 3D modelling, medical imaging, industrial inspection, cultural-heritage digitization, and augmented reality.

10. Light at Surfaces — Phong Illumination Model

Definition

The Phong illumination model is a local shading model that approximates the light reflected from a surface using ambient, diffuse, and specular components.

It is commonly used in computer graphics to simulate surface appearance and is relevant to understanding shading cues used in shape estimation.

Components

Ambient reflection: approximates general background illumination.

Diffuse reflection: depends on the angle between the surface normal and light direction.

Specular reflection: represents shiny highlights and depends on the viewing direction and surface shininess.

Mathematical formula

$$I= k_aI_a+ k_dI_l\max(0,\mathbf{n}\cdot\mathbf{l}) +k_sI_l\max(0,\mathbf{r}\cdot\mathbf{v})^m$$

where:

- k_a,k_d,k_s are ambient, diffuse, and specular reflection coefficients.
- I_a and I_l are ambient and light intensities.
- $\mathbf{n}$ is the surface normal.
- $\mathbf{l}$ is the light direction.
- $\mathbf{r}$ is the reflected light direction.
- $\mathbf{v}$ is the viewing direction.
- m controls the sharpness of the highlight.

Block diagram

Light Source + Surface + Viewer

Ambient

Diffuse

Specular

Combine the Three Components

Final Surface Intensity

Working

1. Define the light source and viewing direction.
2. Determine the surface normal.
3. Calculate ambient contribution.

4. Calculate diffuse reflection using the normal-light angle.
5. Calculate specular reflection using the reflected direction and viewer.
6. Add the contributions to obtain the final intensity.

Advantages

- Simple and computationally efficient.
- Models diffuse and shiny surfaces.
- Provides a useful approximation for rendering.
- Helps explain the relationship between surface orientation and brightness.

Disadvantages

- It is an empirical local model rather than a complete physical illumination model.
- Does not fully represent global illumination or inter-reflections.
- Specular highlights can complicate shape-from-shading estimation.
- Real surfaces may not follow the assumed reflectance behaviour.

Applications

Computer graphics, surface rendering, virtual environments, shading analysis, and illumination-based shape estimation.

11. Reflectance Map

Definition

A reflectance map describes the relationship between surface orientation and the intensity observed under a specified illumination and reflectance model.

It maps a surface normal, often parameterized by two orientation variables, to a predicted image intensity.

Principle

For a Lambertian surface under fixed illumination:

$$ R(\mathbf{n})=\rho\max(0,\mathbf{n}\cdot\mathbf{l}) $$

For a more complex model, the reflectance map may include ambient and specular components.

Block diagram

Specify Illumination and Material

Choose Surface Normal

Evaluate Reflectance Model

Calculate Predicted Intensity

Build Reflectance Map

Compare with Observed Image

Working

1. Specify a lighting configuration.
2. Select a reflectance model.
3. Evaluate the predicted brightness for possible surface normals.
4. Store the intensity associated with each orientation.
5. Compare observed brightness against the map.
6. Estimate compatible surface orientations.

Advantages

- Provides a mathematical link between shading and orientation.
- Can support shape-from-shading algorithms.
- Accommodates different reflectance models.
- Helps analyse orientation ambiguity.

Disadvantages

- Depends on correct illumination and material assumptions.
- Different normals may produce similar intensities.
- Noise and specular reflections affect matching.
- A reflectance map alone does not provide depth.

Applications

Shape from shading, photometric analysis, surface reconstruction, and computer graphics.

12. Albedo Estimation

Definition

Albedo is a measure of how much incident light a surface reflects. Albedo estimation attempts to determine the intrinsic reflectance of a surface separately from illumination and surface orientation.

For example, a dark object under strong light may produce the same image intensity as a light object under weak light. Separating albedo from shading helps resolve this ambiguity.

Mathematical principle

Under a simplified Lambertian model:

$$ I(x,y)=\rho(x,y)S(x,y) $$

where $\rho(x,y)$ is albedo and $S(x,y)$ is the shading term.

If shading is estimated and is nonzero:

$$ \rho(x,y)\approx\frac{I(x,y)}{S(x,y)} $$

Block diagram

Input Image

Estimate Lighting and Surface Orientation

Calculate Shading Component

Separate Reflectance from Shading

Estimate Albedo Map

Analyse Material / Surface Properties

Working

1. Acquire an image.
2. Estimate the lighting and shading model.
3. Estimate or constrain surface orientation.
4. Separate illumination effects from reflectance.
5. Compute or refine the albedo map.
6. Evaluate material or surface differences.

Advantages

- Helps distinguish material changes from lighting changes.
- Supports intrinsic image decomposition.
- Improves shape-from-shading reliability when albedo is known or estimated.
- Useful for material analysis.

Disadvantages

- Albedo and illumination are difficult to separate from one image.
- Shadows and specular reflections complicate estimation.
- Requires assumptions or additional observations.
- Errors in the shading estimate affect the albedo map.

Applications

Material recognition, intrinsic image decomposition, surface reconstruction, image relighting, and scene understanding.

13. Photometric Stereo

Definition

Photometric stereo estimates the surface normals of an object from multiple images captured from the same camera viewpoint but under different known lighting directions.

Because the viewpoint remains fixed, differences in pixel intensity mainly provide information about surface orientation and reflectance.

Principle

For a Lambertian surface, the intensity under light i is:

$$ I_i=\rho\,\mathbf{n}\cdot\mathbf{l}_i $$

For three or more independent lighting directions, the equations can be combined:

$$ \mathbf{I}=L\mathbf{g} $$

where $\mathbf{g}=\rho\mathbf{n}$. If the lighting matrix L has sufficient rank, estimate:

$$ \mathbf{g}=L^{+}\mathbf{I} $$

Here, L^{+} is the pseudoinverse of L. Then:

$$ \rho=\|\mathbf{g}\|,\qquad \mathbf{n}=\frac{\mathbf{g}}{\|\mathbf{g}\|} $$

These formulas assume the appropriate lighting normalization and an ideal Lambertian surface.

Block diagram

Fixed Camera and Object

Capture Multiple Images Under Different Lights

Combine Intensity Measurements

Estimate Surface Normals and Albedo

Integrate Normals to Recover Surface Shape

Working

1. Keep the camera and object fixed.
2. Illuminate the object from several known directions.
3. Capture an image for each lighting direction.
4. Match the same pixels across all images.
5. Solve the reflectance equations for each pixel.
6. Estimate surface normals and albedo.
7. Integrate the normals to recover depth or surface shape.

Advantages

- Recovers fine surface details.
- Can estimate shape using a fixed camera.
- Separates orientation and albedo under suitable assumptions.
- Useful for surfaces with subtle geometric details.

Disadvantages

- Requires multiple images and controlled illumination.
- Object movement between captures causes errors.
- Shadows and specular reflections violate simple assumptions.
- Integrating normals into depth can accumulate errors.

Applications

Industrial surface inspection, coin and engraving reconstruction, cultural-heritage digitization, material analysis, and fine surface measurement.

14. Surface Smoothness Constraint

Definition

A surface smoothness constraint is a mathematical assumption that neighbouring surface points usually have similar orientations or depths, except at genuine discontinuities such as object boundaries.

It helps regularize shape reconstruction when the image measurements are noisy or incomplete.

Mathematical formulation

For a depth map $z(x,y)$, a simple smoothness penalty is:

$$ E_s=\iint\left[\left(\frac{\partial z}{\partial x}\right)^2+ \left(\frac{\partial z}{\partial y}\right)^2 \right]dx\,dy $$

A common reconstruction objective combines image consistency and smoothness:

$$ E=E_{\text{data}}+\lambda E_s $$

Here, E_{data} measures disagreement with observed image evidence, and λ controls the strength of the smoothness penalty.

Block diagram

Noisy or Incomplete Depth / Normal Estimates

Define Data Consistency Term

Define Surface Smoothness Term

Combine Terms into an Objective

Optimize Surface Estimate

Smooth and Consistent Shape

Working

1. Obtain initial depth or normal estimates.
2. Define how well the estimates fit the image evidence.
3. Add a smoothness penalty for abrupt changes.
4. Combine the terms into an objective function.
5. Minimize the objective using an appropriate optimization method.
6. Recover the refined surface.

Advantages

- Reduces noise and irregular estimates.
- Helps reconstruct incomplete surfaces.
- Provides stable solutions when image evidence is ambiguous.

Disadvantages

- Excessive smoothing removes sharp edges and fine details.
- Incorrect regularization can distort the actual shape.
- The choice of λ affects the result.

Applications

Shape from shading, photometric stereo, depth estimation, 3D reconstruction, and surface denoising.

15. Shape from Texture

Definition

Shape from texture is a technique for estimating the 3D shape or surface orientation of an object by analysing changes in the texture pattern across an image.

When a textured surface is viewed in perspective, texture elements often appear smaller, denser, or more compressed as the surface extends away from the viewer.

For example, floor tiles appear larger near the camera and progressively smaller in the distance.

Principle

Texture provides geometric cues such as:

- Texture gradient: changes in texture size or density.
- Foreshortening: texture elements appear compressed along certain directions.
- Perspective distortion: repeated patterns change with surface orientation.
- Texture orientation: the apparent direction of texture elements changes across a surface.

Block diagram

Input Textured Image

Detect Texture Elements or Patterns

Measure Texture Size, Density and Orientation

Analyse Perspective Distortion

Estimate Surface Orientation / Depth

Reconstruct Surface Shape

Working

1. Capture an image containing a visible texture.
2. Identify repeated texture elements or local texture statistics.
3. Measure variations in element size, spacing, and orientation.
4. Analyse how perspective changes the texture pattern.
5. Estimate surface orientation and relative depth.
6. Combine the estimates to reconstruct the surface.

Advantages

- Can infer shape from a single image.
- Does not always require special lighting.
- Works well on surfaces with regular, identifiable textures.
- Can estimate surface orientation over large regions.

Disadvantages

- Performs poorly on textureless surfaces.
- Repetitive patterns may create ambiguity.
- Irregular textures make geometric estimation difficult.
- Perspective and lighting changes can confuse texture measurements.

Applications

Terrain reconstruction, road-surface analysis, agricultural fields, surface inspection, and 3D scene understanding.

16. Shape from Colour

Definition

Shape from colour estimates surface geometry by analysing spatial changes in colour and the way different surface orientations, materials, and illumination conditions affect observed colour.

Colour can provide useful shape cues when the scene contains known materials, colour gradients, calibrated illumination, or colour patterns associated with geometry.

Important: Colour alone does not uniquely determine 3D shape. Reliable reconstruction requires suitable assumptions or additional information.

Principle

An observed colour can be modelled as a function of material reflectance, illumination, and surface orientation:

$$ C(x,y)=f(\rho_c,\mathbf{n},\mathbf{l}) $$

where C is the observed colour, ρ_c represents the colour-dependent surface reflectance, $\mathbf{n}$ is the surface normal, and $\mathbf{l}$ is the illumination direction.

Block diagram

Input Colour Image

Extract Colour Features

Model Material Reflectance and Illumination

Analyse Colour Variations

Estimate Surface Orientation / Depth

3D Shape Reconstruction

Working

1. Acquire a colour image.
2. Analyse colour distributions and spatial variations.
3. Estimate or calibrate illumination and surface reflectance.
4. Relate colour observations to possible surface orientations.
5. Combine the colour cues with geometric constraints.
6. Recover a surface or depth estimate.

Advantages

- Provides additional information beyond grayscale intensity.
- Can distinguish regions with different materials.
- May complement shading and texture cues.
- Useful when colour patterns correlate with geometry.

Disadvantages

- Colour changes may result from different materials rather than shape.
- White balance and illumination affect the observed colours.
- Shadows and reflections can mislead estimation.

- Shape recovery is ambiguous without further assumptions.

Applications

Colour-based scene reconstruction, material-aware surface analysis, agricultural imaging, medical imaging, and object recognition.

17. Shape from Motion

Definition

Shape from motion is a computer vision technique that estimates 3D structure by observing how an object or scene changes position across multiple images taken from different viewpoints.

As a camera moves, points at different depths shift by different amounts in the image. This motion provides information about the relative 3D structure of the scene.

Two important related techniques are Structure from Motion (SfM) and motion-based depth estimation.

Principle

For a moving camera, the apparent image displacement depends on camera movement, scene depth, and viewing geometry. Multiple views provide constraints on 3D point positions and camera poses.

Block diagram

Capture Images from Different Viewpoints

Detect Distinctive Features

Match Features Across Images

Estimate Camera Motion

Triangulate 3D Feature Points

Optimize Camera Poses and 3D Points

Sparse / Dense 3D Reconstruction

Working

1. Capture several overlapping images from different viewpoints.
2. Detect distinctive features such as SIFT keypoints.
3. Match corresponding features between images.
4. Estimate relative camera poses using epipolar geometry.

5. Triangulate matched points to obtain 3D coordinates.
6. Refine the camera poses and 3D points using bundle adjustment.
7. Produce a sparse or dense reconstruction.

Advantages

- Recovers 3D structure from ordinary images.
- Can work without a dedicated depth sensor.
- Supports large-scale scene reconstruction.
- Multiple viewpoints help resolve ambiguities present in a single image.

Disadvantages

- Requires sufficient viewpoint variation and overlapping features.
- Repetitive patterns and textureless areas are difficult.
- Moving objects can violate static-scene assumptions.
- Scale is generally ambiguous in monocular reconstruction unless an external scale reference is available.

Applications

3D mapping, robotics, drone surveying, cultural-heritage reconstruction, augmented reality, and autonomous navigation.

18. Shape from Edges

Definition

Shape from edges estimates the geometry or structure of an object by analysing visible image boundaries, contours, and line features.

Edges reveal changes in intensity and can indicate object silhouettes, surface discontinuities, creases, and intersections of surfaces.

Principle

The geometry of image contours provides constraints on possible 3D shapes. For example, a silhouette restricts the object to lie within a viewing cone defined by the camera and the contour.

With multiple silhouettes from different viewpoints, their intersection can estimate the object's visual hull.

Block diagram

Input Image(s)

Edge / Contour Detection

Link Edges and Extract Silhouettes

Apply Geometric Constraints

Combine Multiple Views if Available

Estimate 3D Shape

Working

1. Detect edges using Canny, Sobel, or another edge detector.
2. Connect edge fragments into contours.
3. Identify object silhouettes and meaningful line structures.
4. Apply geometric constraints based on camera calibration and contour information.
5. If multiple views are available, combine their silhouette constraints.
6. Estimate the object's shape or boundary structure.

Advantages

- Uses clear object boundaries.
- Works well for objects with distinctive silhouettes.
- Can reconstruct visual hulls from multiple views.
- Does not require detailed texture on the object surface.

Disadvantages

- Weak or broken edges cause incorrect contours.
- A single silhouette cannot recover all hidden surface details.
- Concave regions may be missed by visual-hull reconstruction.
- Internal texture and material properties are not directly recovered from silhouettes.

Applications

Industrial shape measurement, object modelling, silhouette-based reconstruction, robotics, and 3D animation.